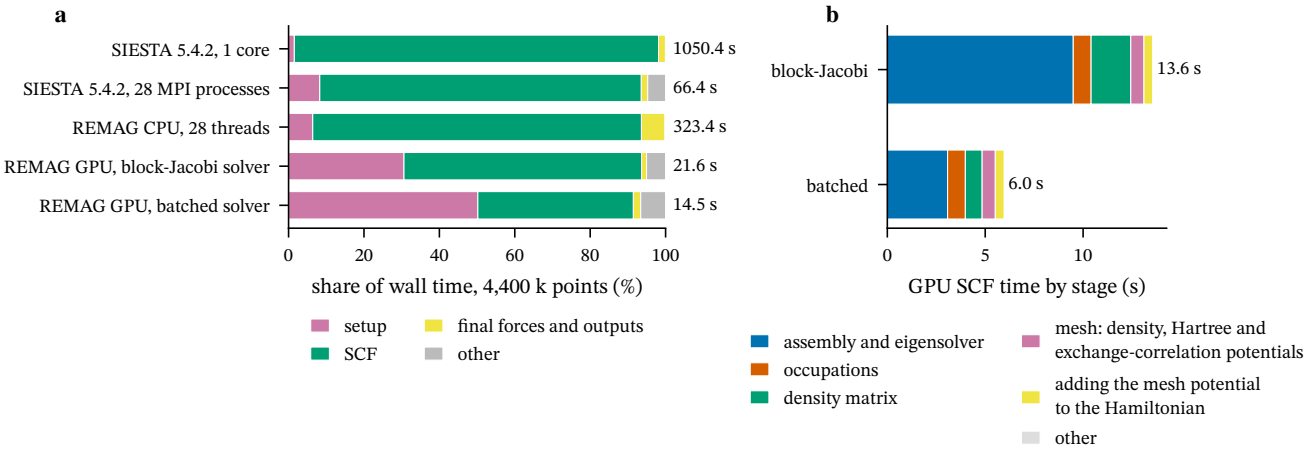
a
SIESTA 5.4.2, 1 core
SIESTA 5.4.2, 28 MPI processes
REMAG CPU, 28 threads
REMAG GPU, block-Jacobi solver
REMAG GPU, batched solver
1050.4 s
66.4 s
323.4 s
21.6 s
14.5 s
0
20
40
60
80
100
share of wall time, 4,400 k points (%)
setup
SCF
final forces and outputs
other
b
block-Jacobi
batched
13.6 s
6.0 s
0
5
10
GPU SCF time by stage (s)
assembly and eigensolver
occupations
density matrix
mesh: density, Hartree and exchange-correlation potentials
adding the mesh potential to the Hamiltonian
other